

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 12 (Chapter 7 and some parts of Chapter 2 and 9)

The Medium Run: A model for the labor market

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WHERE WE ARE

Fundamentals

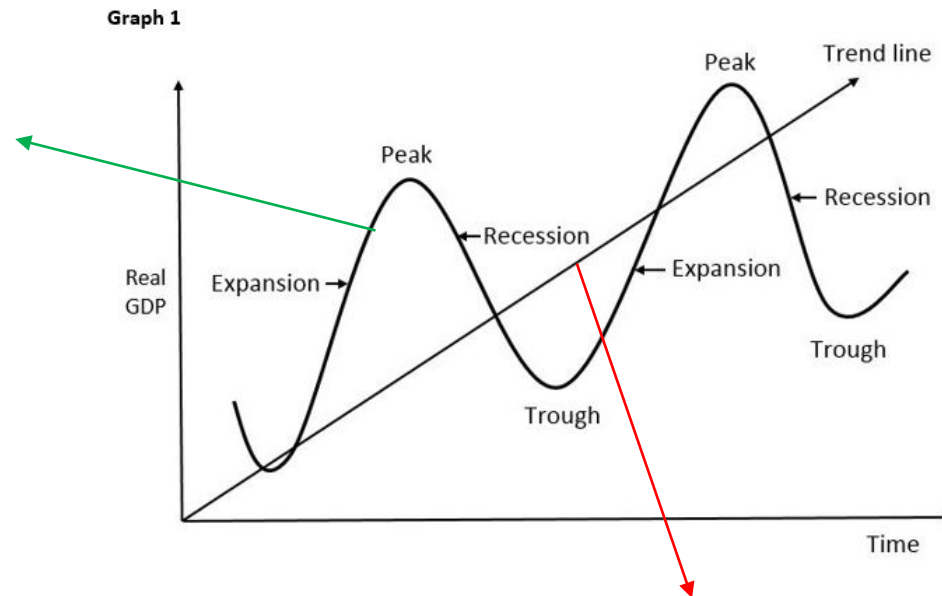
1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

WHAT DETERMINES THE GDP?

Short run
GDP determined by
demand factors
Actual output



Medium run
GDP determined by supply factors
Potential GDP (Natural GDP)

SHORT RUN

- Rigid prices
- Price expectations are not necessarily correct: **Expected prices $\neq$ actual prices (observed)**
- **Actual output** is different from the potential
- **Output gap** = difference between actual GDP and estimated potential GDP

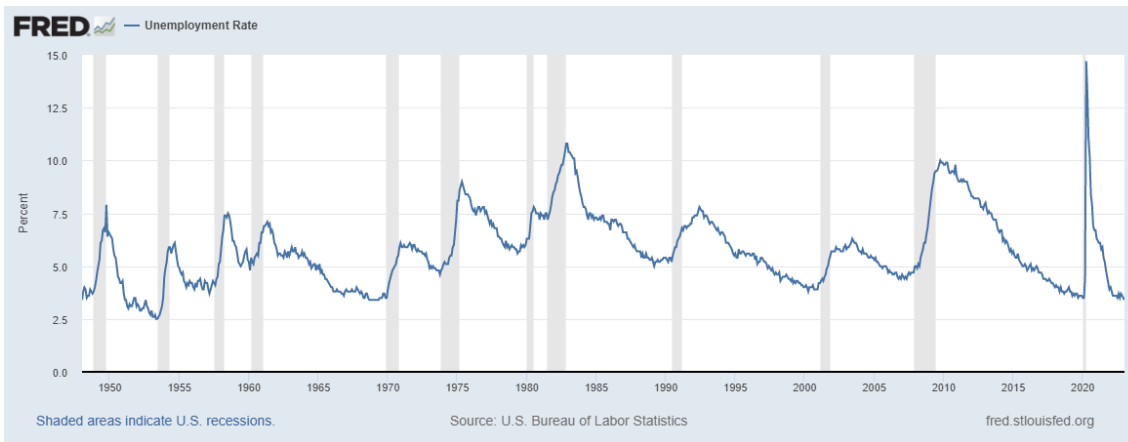
MEDIUM RUN

- Prices change
- Expectations are correct: **Expected prices = actual prices**
- **Potential output** (natural level) = full employment GDP («*normal*» use of factors)
- The economy returns to the level of production associated to the **natural rate of unemployment**

Transition from short to medium run:

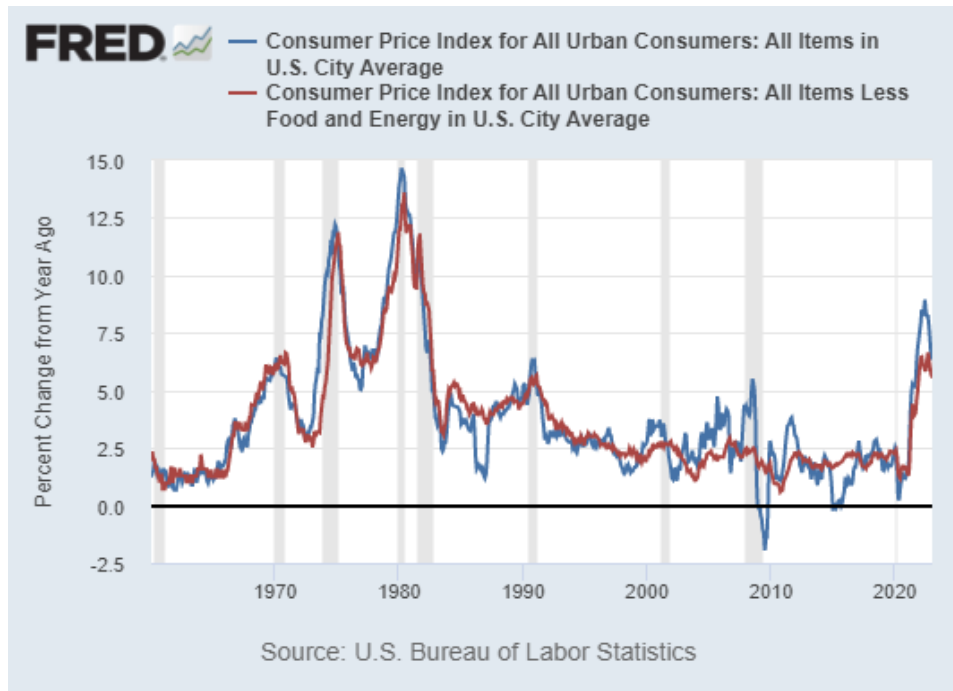
Prices and wages adjust over time affecting the amount an economy can produce => main ingredients

1. The labor market
2. Price adjustments
3. Expectations



Unemployment increases during recessions and decreases during expansions

Source: Series UNRATE: Federal Reserve Economic Data (FRED), <http://research.stlouisfed.org/fred2/>



The rate of inflation peaks at the end of an expansion and falls during (after) a recession

FROM THE REAL ECONOMY TO PRICE DYNAMICS

If $\uparrow Z$, to produce more we need more workers

$\Rightarrow$ $\downarrow$ unemployment and $\uparrow$ employment

$\Rightarrow$ workers have higher bargain power

$\Rightarrow$ $\uparrow$ wages

$\Rightarrow$ $\uparrow$ firms' production costs

$\Rightarrow$ $\uparrow$ prices

To understand this mechanism
we need to study the bargaining
process btw firms and workers in
the labor market

THE LABOR MARKET

Market where wages are determined based on the bargaining process
btw firms and workers

Total population

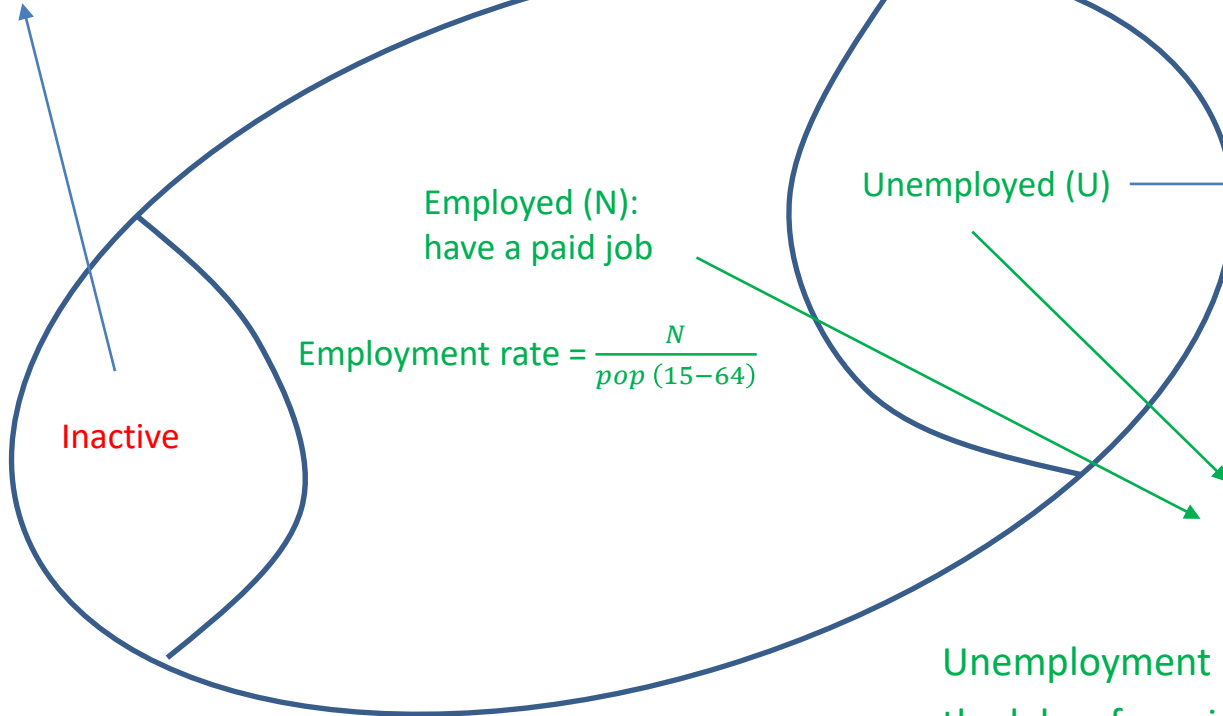


Working age population

People aged 15 to 64



- Do not have a job
- Not looking for a job
- Not willing to work
- => out of the labor force



- Do not have a job
- Are looking for a job
- Are willing to work

$$\text{Employment rate} = \frac{N}{\text{pop (15-64)}}$$

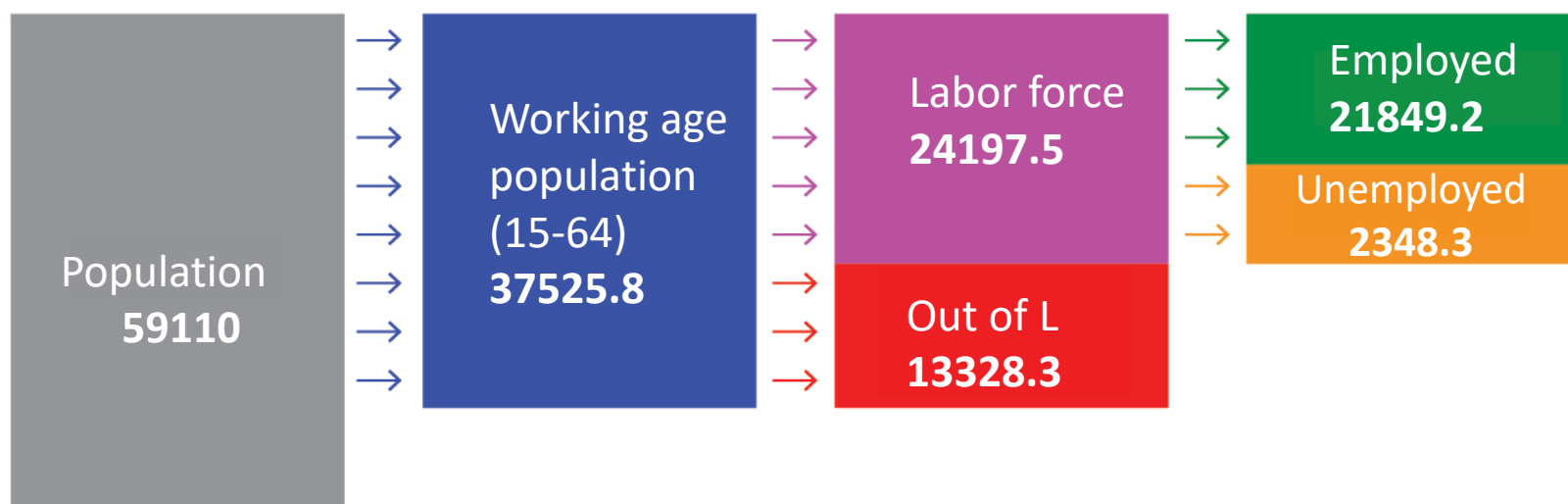
$$\text{Labor force} = N + U$$

$$\text{Unemployment rate} = u = \frac{U}{L} \text{ share of the labor force in unemployment}$$

$$\text{Participation rate} = \frac{L}{\text{pop (15-64)}}$$

LABOR MARKET

Italy (2021) *thousands*



Participation rate = 64.5%

Employment rate (N/working age population) = 58.2%

Unemployment rate = 9.7%

NOTE ON UNEMPLOYMENT

To describe the situation in the labor market, we often use the unemployment rate ...

...BUT...

...this is not enough to give up a complete picture of the labor market, it is important to consider also the participation rate and the number of discouraged workers

- **Example:**

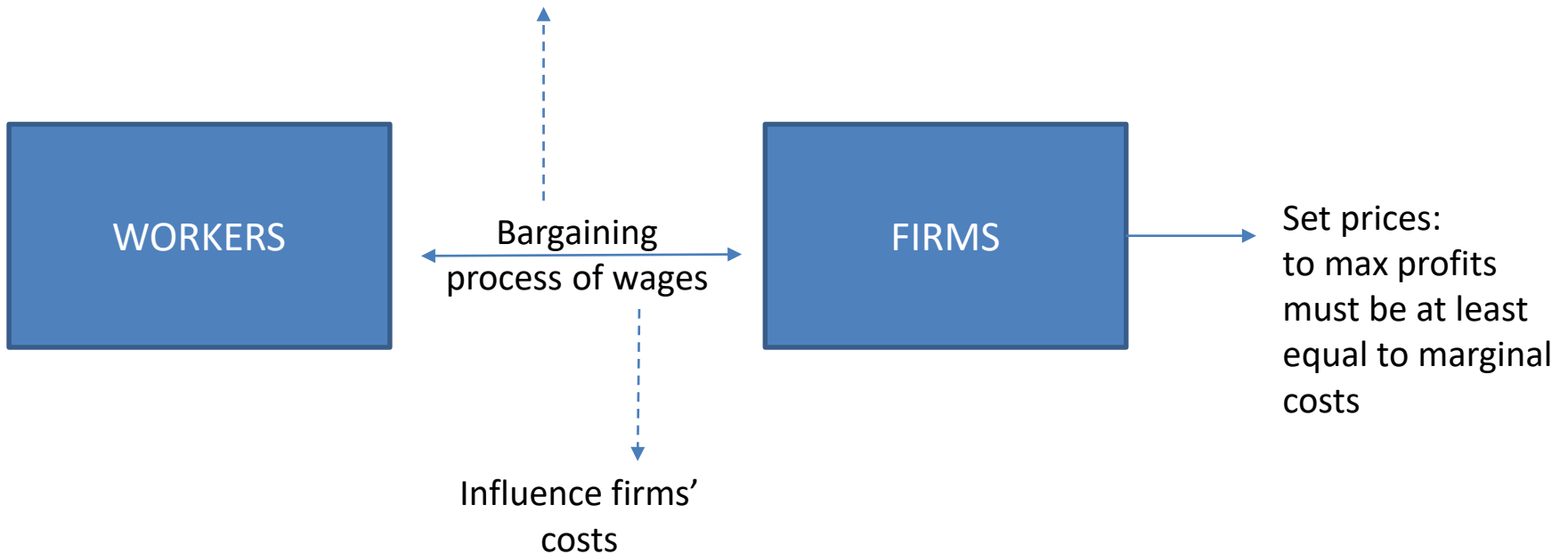
Initial situation in a country: $N = 900$, $U = 100$

Two scenarios:

1. CASE A: $\Delta U = -10$; $\Delta N = +10 \Rightarrow u = \frac{90}{1000} = 9\%$

2. CASE B: $\Delta U = -10$; $\Delta N = 0 \Rightarrow u = \frac{90}{990} = 9,1\%$

The final level of wages depends
on the relative bargaining power
of firms wrt workers



LABOR MARKET MODEL

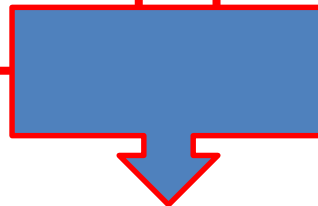
Let's build a model to represent the labor market and the determinants of
real wages (W/P) and **unemployment (u)**

WAGE SETTING (WS)

Equation describing **wages requested by workers** in the bargaining process

PRICE SETTING (PS)

Equation describing **wages that firms are willing to pay** based on the prices they set for their products



Labor market equilibrium

To build a model that describes how the labor market works, we need to observe the characteristics of labor markets and understand:

- 1) what wage workers are willing to accept
- 2) what wage firms are willing to pay
- 3) what price level is set by firms

WHAT ARE THE DETERMINANTS OF WAGES?

- **Conditions prevailing on the labor market:** How simple is to find a job
- **Expected Prices:** We care about our purchasing power
- **Bargaining power of workers** relative to firms that depends on
 - Collective vs. individual bargaining
 - Degree of unionization
 - Cost of substituting the worker
 - Minimum wage
 - Unemployment benefits
- Workers have a **reservation wage** making them indifferent between working and not working
- Usually, workers earn a **wage above their reservation wage**
- Firms pay **efficiency wages:** wages higher than the reservation one to
 - Increase productivity
 - Reduce shirking
 - Reduce turnover

WAGE SETTING

Bargaining process btw workers and firms for the *NOMINAL WAGE (W)*

The nominal average wage level in the economy depends on

1. Expected price level
2. Labor market conditions
3. Labor market institutions

1. Expected prices:

Workers care about the **purchasing power** of their nominal wages (real wages)

Wages set in nominal terms when the level of prices is unknown (long-term contracts)

When workers expect higher prices ask for higher nominal wages, and if firms expect to sell at a higher price are willing to pay more their workers

=> positive relationship btw expected prices and nominal wages

$$P^e \uparrow \Rightarrow W \uparrow$$

2. Labor market conditions:

The bargaining process depends on how difficult is for workers (firms) to find a new job (worker)

When unemployment is high:

is more difficult for workers to find a job since many other workers are looking for a job

is more simple for firms to find workers

=> The bargaining power of workers decreases and that of firms increases

=> bargained wage will be lower

=> Negative relationship btw unemployment rate and nominal wages

$$u \uparrow \Rightarrow W \downarrow$$

3. Labor market institutions:

Some labor market features affect positively workers bargaining power

- Unemployment benefits (level and duration)
- Minimum wage
- High Employment protection (EPL)
- Strong unions
- High firing costs
- ...

→ z = labor market institutions
that give more power to
worker and less to firms

=> Positive relationship btw these structural features of the labor market and nominal wages

$$z \uparrow \Rightarrow W \uparrow$$

In a synthetic way the nominal aggregate wage level in the economy will be

$$W = P^e F(u, z)$$

- +



Wage equation:

explains the determinants of the average
nominal wage level bargained btw firms
and workers

PRICE SETTING

What are the determinants of the average price level in the economy (P)?
How do firms set prices?

Firms want to max profits

Prices set by firms depend on costs

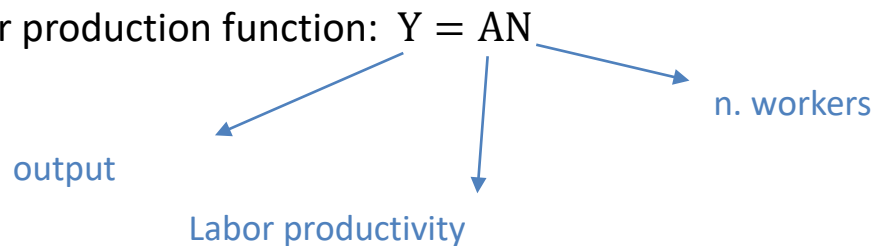
Costs depends on **technology** and input costs



Production function: relationship
between inputs and output

ASSUMPTIONS

- Assumption: only 1 factor – Labor, N
- $L = N + U$ = population $\Rightarrow$ no inactive in the economy
- $u = \frac{U}{L} = \frac{L-N}{L} = 1 - \left(\frac{N}{L} \right) \rightarrow$ Employment rate



(Your book simplifies assuming $A = 1 \Rightarrow Y = N$)

Firms set prices to cover their production costs

How much does it cost to produce 1 unit?

- Using the production function, we calculate labor demand as a function of the quantity produced:

$$Y = AN \Rightarrow N = \frac{Y}{A} \longrightarrow \text{n. Workers necessary to produce } Y$$

- Total production cost:

$$TC = W \times N = W \frac{Y}{A}$$

- Marginal cost of producing 1 unit :

$$MC = \frac{W}{A} = \frac{1}{A} W$$

n. workers

necessary to produce 1 unit

Cost of 1 worker

If firms have some **market power** would charge a price higher than the marginal cost

$$p > MC$$

They charge a mark up over the marginal costs

$$P = MC + m MC = (1 + m)MC$$

markup over the marginal cost

Substituting the marginal cost the price level will be

$$P = (1 + m) \frac{W}{A}$$

Price equation: rule
used to set prices

Given the price setting rule

$$P = (1 + m) \frac{W}{A}$$

- If $m \uparrow \Rightarrow P \uparrow$:
 - Firms have higher market power => charge higher prices
- If $W \uparrow \Rightarrow P \uparrow$:
 - Higher wages => higher production costs => higher prices
- If $A \uparrow \Rightarrow P \downarrow$:
 - Higher productivity => less workers needed to produce 1 additional => lower production costs => lower prices

What is behind m ?

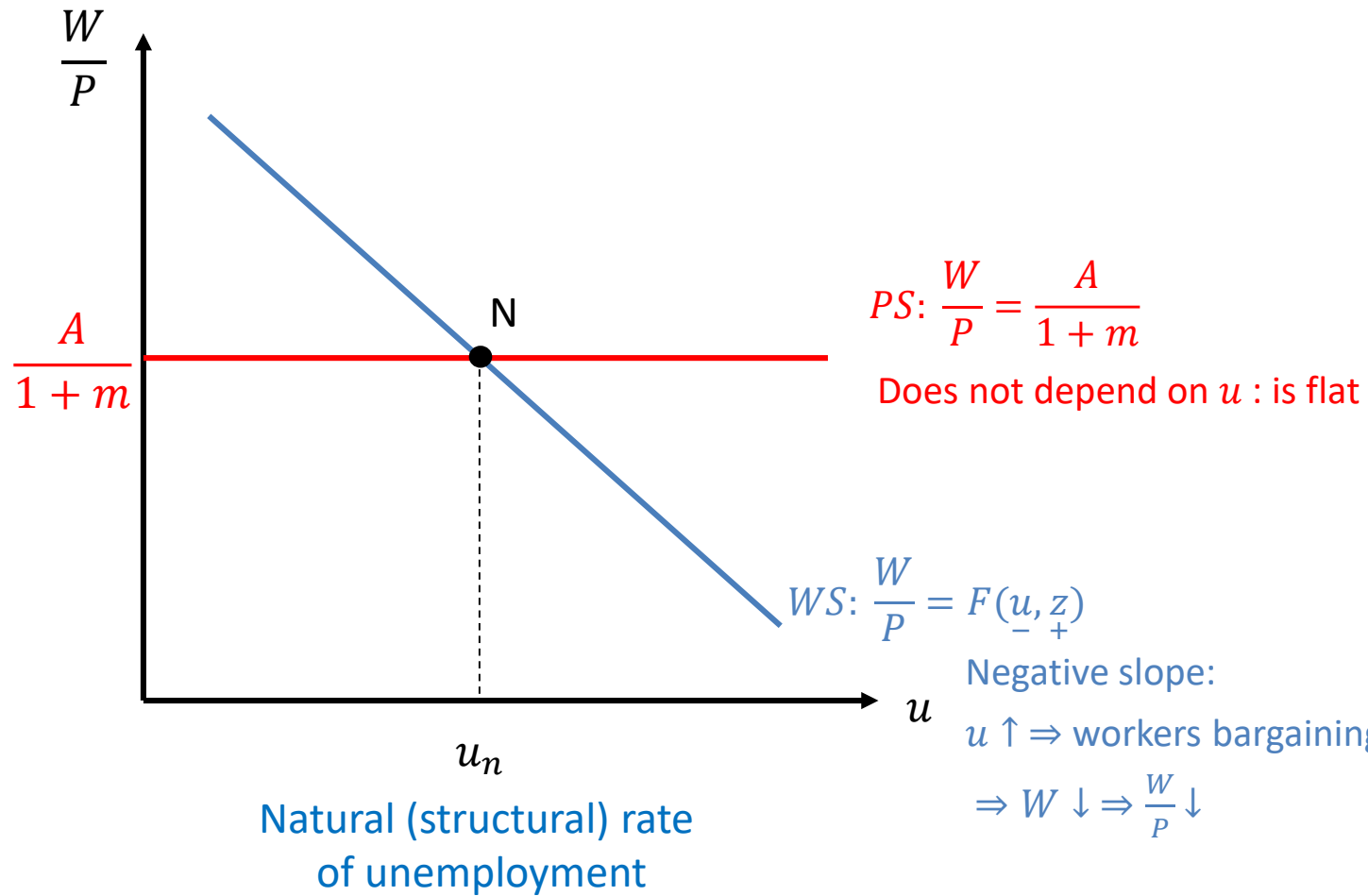
- m is a measure of firms' market power (it $\uparrow$ when firms have higher market power)
 - Structural/institutional features of the goods market
 - Competition level
 - Entry barriers
 - Regulation
 - Anti trust norms
 - ...
- In a perfectly competitive market $m = 0 \Rightarrow P = MC$

MEDIUM RUN EQUILIBRIUM

The real wage and the unemployment rate are at a level that is consistent with the wage-setting and price-setting rules, and expectations are correct

$$\left\{ \begin{array}{l} W = P^e F(u, z) \\ P = (1 + m) \frac{W}{A} \\ P = P^e \end{array} \right. \quad \left\{ \begin{array}{l} W = PF(u, z) \\ P = (1 + m) \frac{W}{A} \end{array} \right. \quad \left\{ \begin{array}{l} W = PF(u, z) \\ \frac{W}{P} = \frac{A}{1 + m} \end{array} \right. \begin{array}{l} \xrightarrow{\text{blue}} \text{WS CURVE} \\ \xrightarrow{\text{red}} \text{PS CURVE} \end{array}$$

Curves plotted in the $(u, W/P)$ plane



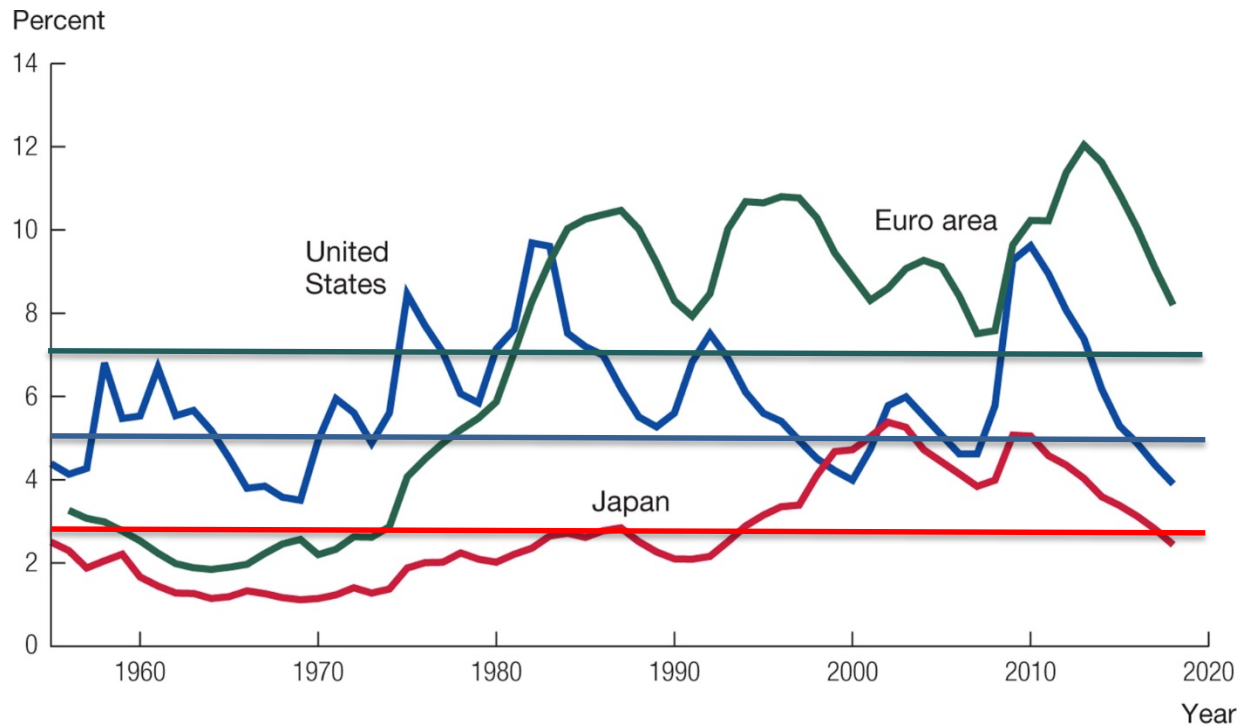
Depends only on *structural factors* of the

- labor market (z)
- Product market (m, A)

Actual unemployment rate (u_t) vs. natural unemployment rate (u_n)

Depends only on the phase of the business cycles (booms/boosts)

Being structural it does not depend on short run shocks in the demand



Source: OECD Main Economic Indicators and the U.S. Bureau of Labor Statistics.
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THE EQUILIBRIUM: INTUITION

Why $u_n > 0$?

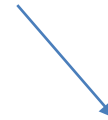
In the market there are **natural frictions** in the job creation process



The «right» match between labor demand and supply takes times



Workers have to find the «best» job for them



Firms have to find the «best» worker to fill their vacancies



Also in equilibrium there is some *mismatch btw labor demand and supply*:
the market forces cannot eliminate completely unemployment.

POTENTIAL OUTPUT

How much does the economy produce in the medium run?

In the labor market

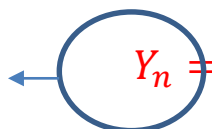
$$N = L - U = L \left(1 - \frac{U}{L} \right)$$
$$N = L(1 - u)$$

In the medium run $u = u_n$

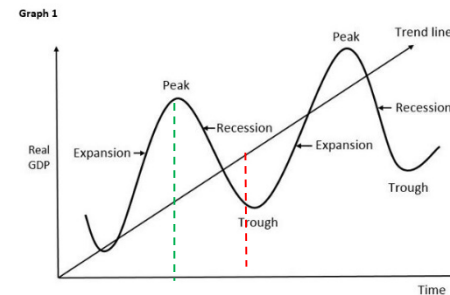
$$N_n = L(1 - u_n)$$

Given the production function $Y = AN$ we obtain:

Potential (natural or
structural) output


$$Y_n = AN_n = AL(1 - u_n)$$

- potential output Y_n
 - medium run level
 - determined only by structural factors of the labor market and the products market
- actual output Y_t
 - short run level
 - determined by demand factors



$$Y_t - Y_n = \text{Output Gap}$$

Positive O.G.
economy above
its potential

Negative O.G.
economy below
its potential

SUMMING UP

- We have built a model to describe the functioning of the labor market considering
 - Workers behavior: wage setting equation
 - Firms behavior: price setting equation
- In the medium run
 - expectation are correct
 - the economy is at its potential
 - real wage required= real wage paid
- The medium run level of unemployment and output does not depend on the demand

IS IT POSSIBLE TO CHANGE u_n AND Y_n ?

Yes, but only with structural changes in the markets

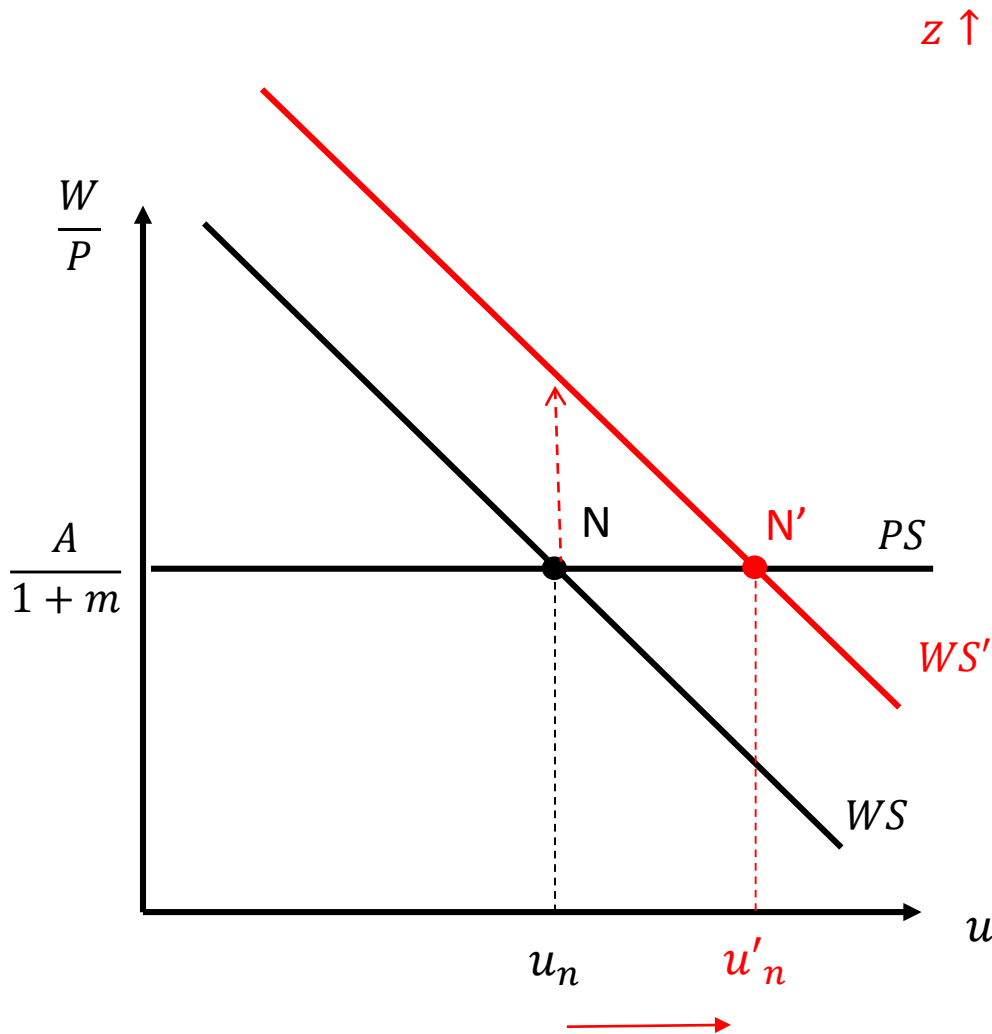
=> structural reforms of the labor market or of the goods market

MORE GENEROUS UNEMPLOYMENT BENEFITS

The government passes a reform to make unemployment benefits more generous (in their amount or in their length)

How can we analyze this shock in the model?

The government passes a reform to make unemployment benefits more generous



$$WS: \frac{W}{P} = F(u, z)$$

$$PS: \frac{W}{P} = \frac{A}{1+m}$$

What happens graphically ?

Workers ask for higher wages: $WS \uparrow$
 PS unchanged

Equilibrium moves from N to N'

In the medium run

- the real wage paid by firms is unchanged
- $u_n \uparrow$ while $Y_n \downarrow$

What happens economically?

What is the economic intuition behind the change in the equilibrium?

- Increase in the reservation wage at the initial unemployment rate
- Workers want higher wages with respect to the one that firms are willing to pay to maximize their profits
 - ↓ search intensity among workers
 - ↓ acceptance rate of offers among unemployed
- More frictions in the matching between demand and supply
- In the medium run the unemployment rate increases and the output decreases
- Nominal wages are higher, production costs increase but real wages paid by firms remain unchanged
- The unemployment rate increases to induce workers to accept the wage paid by firms

INCREASE IN ENTRY BARRIERS IN THE GOODS MARKET

The government introduces a new regulation of the goods market, making it more difficult for incumbents to enter

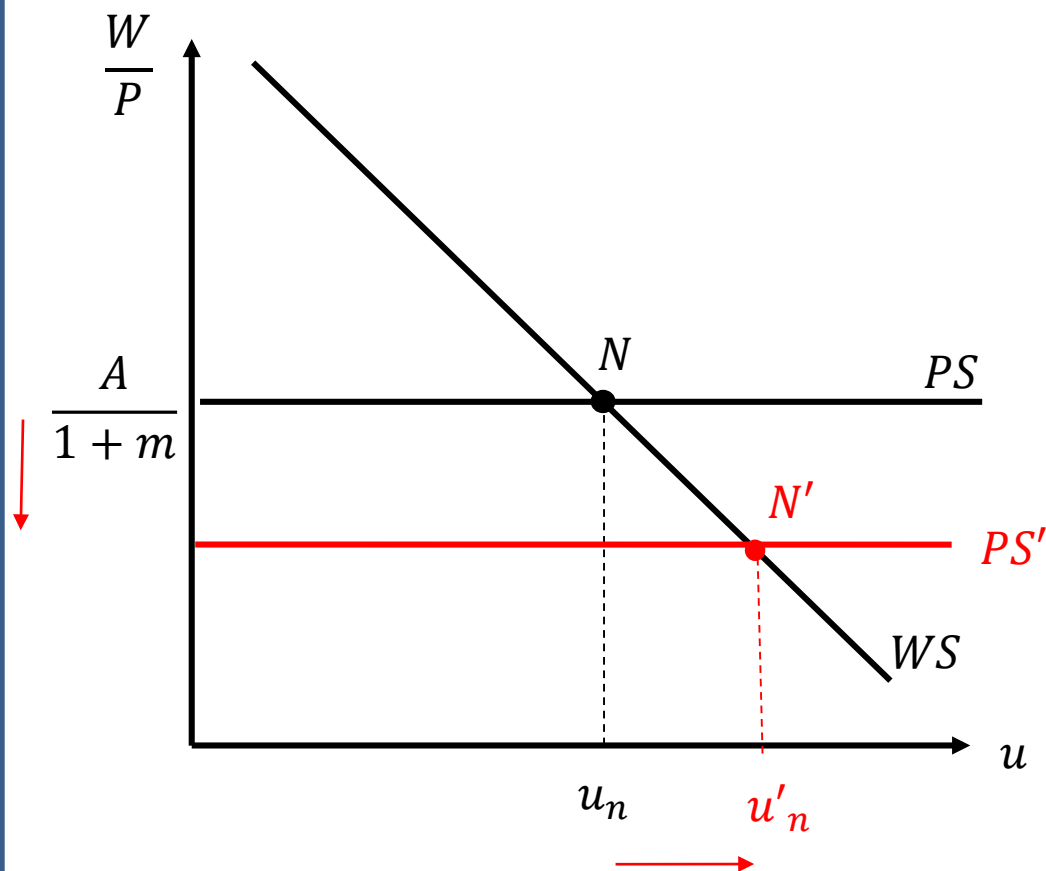
How can we analyze this shock in the model?

After the reform firms will have more market power in the product market

$m \uparrow$

$$WS: \frac{W}{P} = F(u, z)$$

$$PS: \frac{W}{P} = \frac{A}{1+m}$$



What happens graphically ?

Firms set higher prices and the real wage that maximizes profits decreases: $PS \downarrow$
 WS unchanged

Equilibrium moves from N to N'

In the medium run

- the real wage paid by firms $\downarrow$
- $u_n \uparrow$ while $Y_n \downarrow$

What happens economically?

What is the economic intuition behind the change in the equilibrium?

- Decrease in real wages that allow firms to maximize profits at the initial unemployment rate
- Workers want higher wages with respect to the one that firms are willing to pay to maximize their profits
 - ↓ search intensity among unemployed
 - ↓ acceptance rate of offers among unemployed
- More frictions in the matching between demand and supply
- In the medium run the unemployment rate increases and the output decreases
- Price level is higher and real wages paid by firms decrease
- The unemployment rate increases to induce workers to accept the lower real wage paid by firms

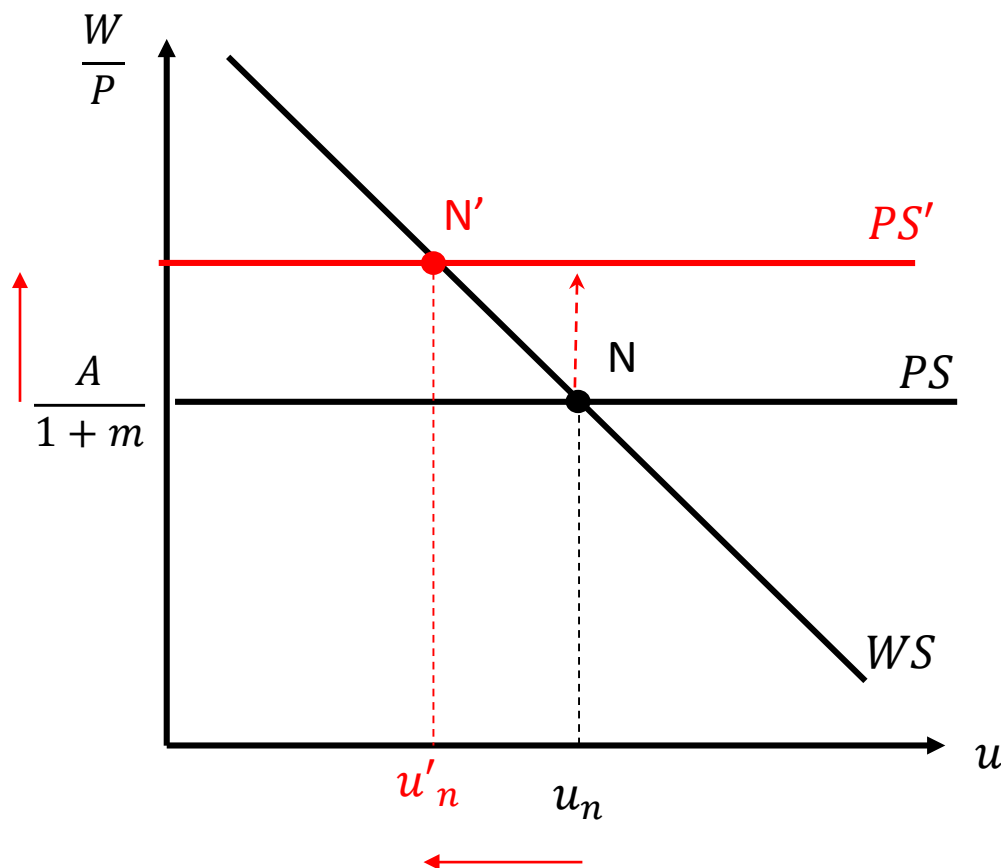
INCREASE IN PRODUCTIVITY

Firms make a huge investment in new information technologies

How can we analyze this shock in the model?

The investment in new information technologies increases productivity

$A \uparrow$



$$WS: \frac{W}{P} = F(u, z_+)$$

$$PS: \frac{W}{P} = \frac{A}{1+m}$$

Which curve is affected?

Firms set lower prices and the real wage that maximizes profits increases:
 $PS \uparrow$

WS unchanged

Equilibrium moves from N to N'

In the medium run

- the real wage paid by firms $\uparrow$
- $u_n \downarrow$ while $Y_n \uparrow$

What happens economically?

What is the economic intuition behind the change in the equilibrium?

- The investment in new IT increases labor productivity
- Since the same workers can produce more, production costs decreases and firms can charge lower prices so that the real wage that maximizes firms profits increases
- Firms are willing to pay higher real wages with respect to the one asked by workers
 - ↑ search intensity among unemployed
 - ↑ acceptance rate of offers among unemployed
- Less frictions in the matching between demand and supply
- In the medium run the unemployment rate decreases and the output increases
- Price level is lower and real wages paid by firms increase

QUICK CHECK

The natural rate of unemployment

- a) does not change when a minimum income scheme is introduced in a country.
- b) increases when markets are highly competitive
- c) **decreases when firms reduce their markups**
- d) is “natural” and therefore independent of the institutional characteristics of the labor market

QUICK CHECK

Suppose the government of your country implements a reform that increases worker protection by increasing firms firing costs. In addition the country, invest public resources in education and training program for workers, which increases their productivity by 25%. In the medium run the unemployment rate _____ and potential output_____.

- a) will increase, will increase
- b) will increase, will decrease
- c) will decrease, will increase
- d) **could not change, could not change**

QUICK CHECK

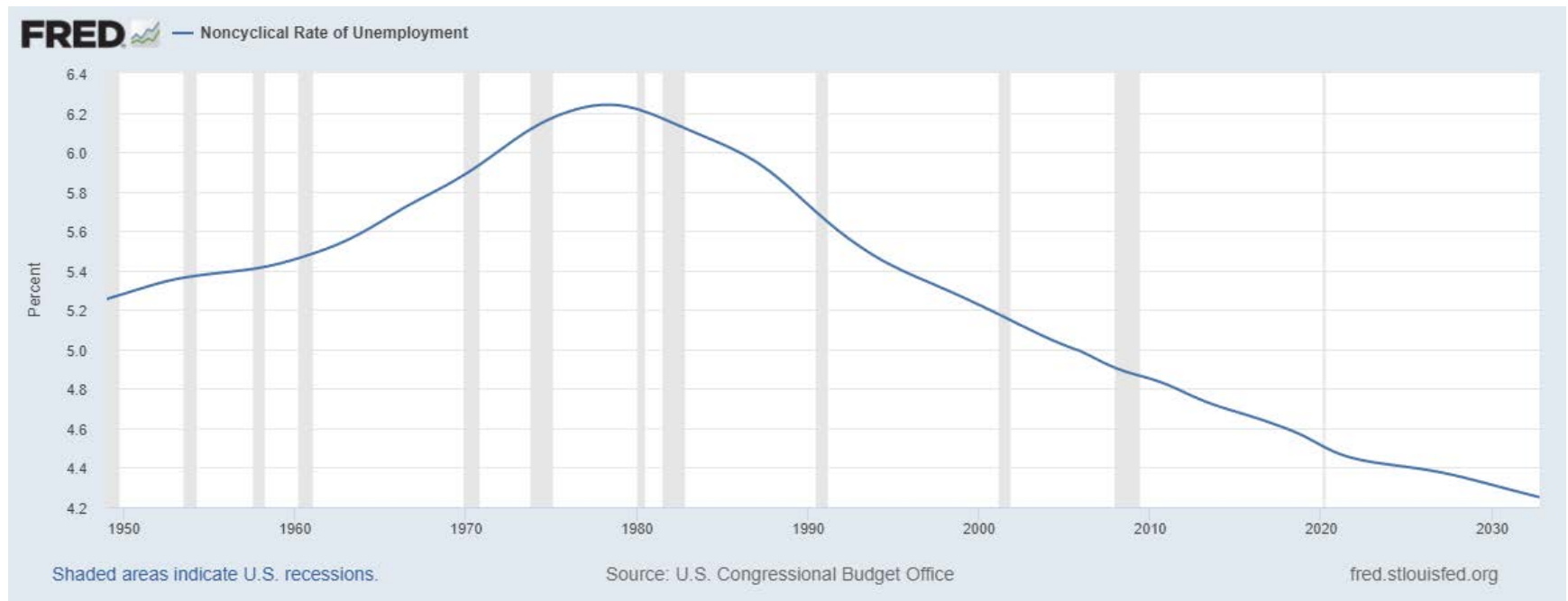
Let's imagine a global energy crisis occurs, leading to a sharp increase in energy prices. Assuming that energy is a fundamental input for production, what is the most likely effect in the medium run?

- a) Real wages will increase because firms will raise prices to cover costs.
- b) Real wages will decrease, and the natural rate of unemployment will increase.**
- c) Real wages will decrease, but the natural rate of unemployment will remain unchanged.
- d) The energy crisis will only have short-term effects, with no consequences in the medium run.

CONCLUSIONS

- Structural differences across countries explain differences in their performance:
Institutions matter and make the difference
- Changing the structure of an economy takes time
- Potential output and unemployment are not observed, they are only estimated

THE NATURAL RATE OF UNEMPLOYMENT- USA



The natural rate of unemployment (NAIRU) is the rate of unemployment arising from all sources except fluctuations in aggregate demand. Estimates of potential GDP are based on the long-term natural rate.

Not observed, estimated!!!

$$u_n \text{ EU} > u_n \text{ US}$$

TAKE AWAY AND KEY CONCEPTS

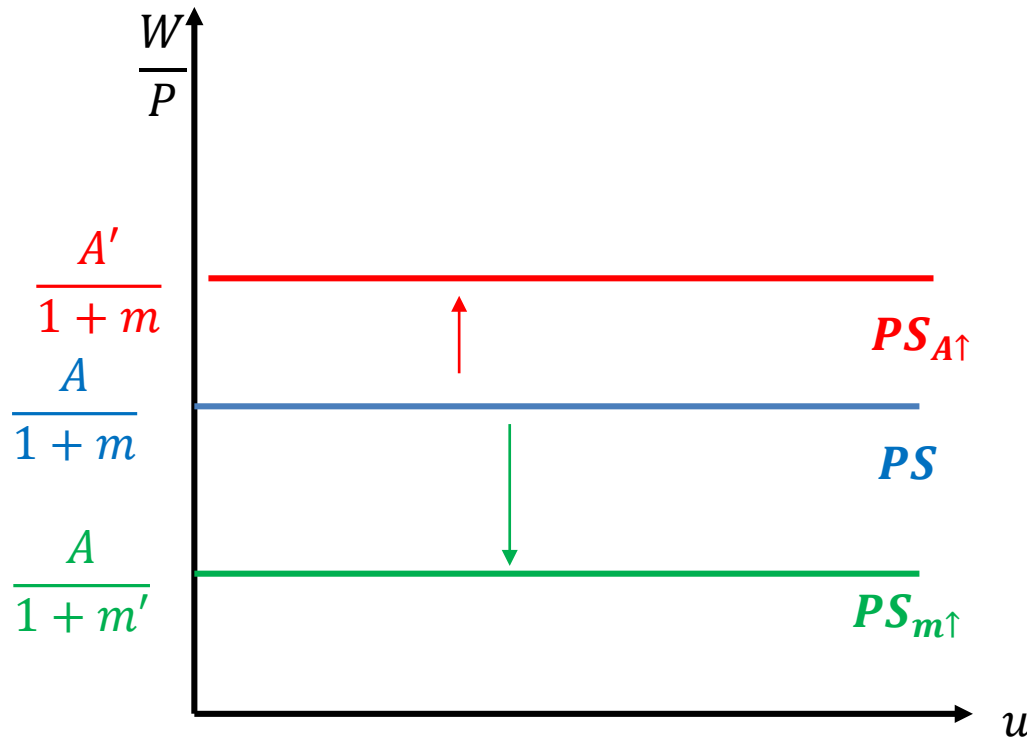
- Labor force
- Unemployment rate
- Participation rate
- Factors affecting the bargaining power of workers and firms
- Wage equation
- Marginal costs
- Wage setting
- Price setting
- Natural unemployment
- Natural output
- Output gap
- Structural shocks

SURVIVAL RECAP

Shifts in the curves

PRICE SETTING CURVE

$$PS: \frac{W}{P} = \frac{A}{1 + m}$$



Does not depend on u

Position depends on m and A

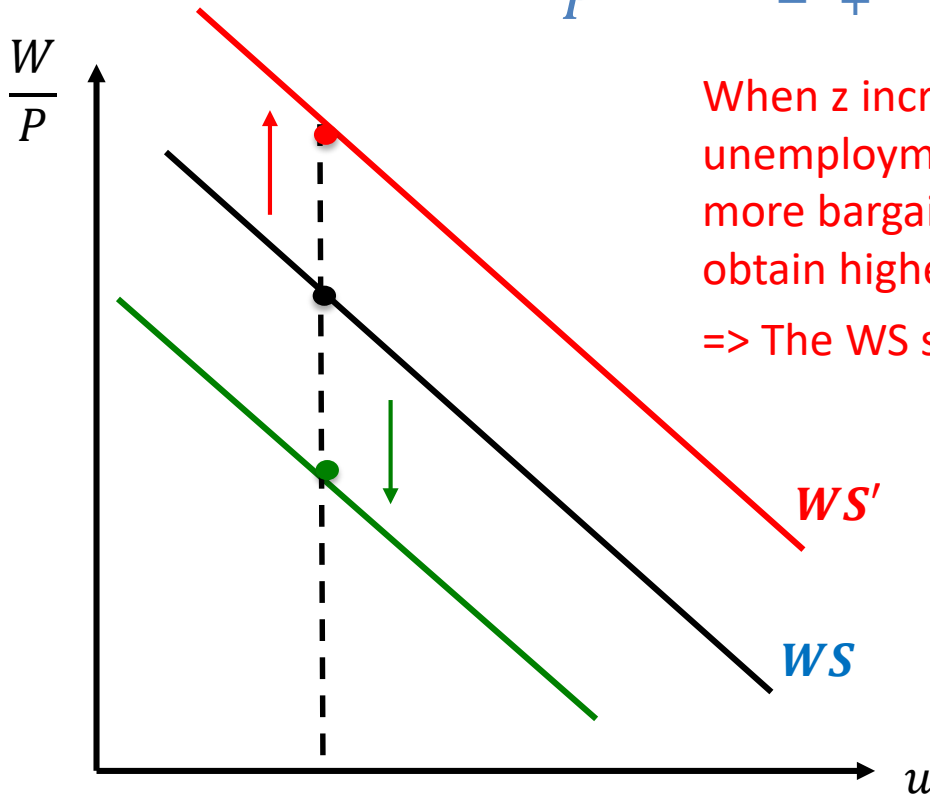
Markup: $m \uparrow \Rightarrow P \uparrow \Rightarrow \frac{W}{P} \downarrow$

Productivity: $A \uparrow \Rightarrow \frac{W}{P} \uparrow$

WAGE SETTING CURVE

$$WS: \frac{W}{P} = F(u, z)$$

Position of the WS depends on z .



When z increases, given the unemployment rate, workers have more bargaining power $\Rightarrow$ ask for and obtain higher nominal and real wages $\Rightarrow$ The WS shifts upward

When z decreases, workers in the economy are less protected, and their bargaining power relative to firms decreases $\Rightarrow$ they ask for and receive lower nominal and real wages $\Rightarrow$ the WS curve shifts downward



**FOR THOSE WHO WANT TO DIVE
DEEPER ...**

THE LABOR MARKET IN THE SHORT RUN

Why in the short run $Y_t \neq Y_n$?

Simply because $P^e \neq P$

- Assume $u_0 < u_n$
 - $\frac{W}{P_0^e} > \frac{W}{P_0} \Rightarrow$ Underestimation of current prices
 - Workers will ask higher nominal wages
 - Given prices that allow to max profits. Firms offer the same real wage as before
 - It is necessary a higher unemployment rate to induce worker to accept the lower W/P
- Over time, expectations adjust and in the medium run

$$P^e = P \Rightarrow u = u_n$$

